

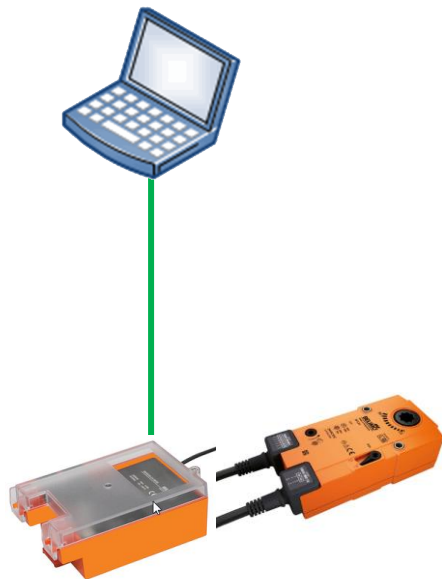
B2 - BACnet / Modbus (Advanced)

Modbus - Workshop I

BKN

B2 - Modbus - Workshop I – BKN - Network

Modbus Poll



BKN

— Serial network (RS485)

B2 - Modbus - Workshop I - BKN

1. Power your device.

 *See wiring diagrams in the datasheet*

2. Connect your BKN via Modbus RTU with your laptop. Connect the actuator to the BKN.

 *Use RS485-USB adapter*

 *See wiring diagram in the datasheet*

3. Configure your actuator according to network overview.

 *Find your appropriate configuration tool in the documents (LINK.10, Belimo Assistant 2, etc.)*

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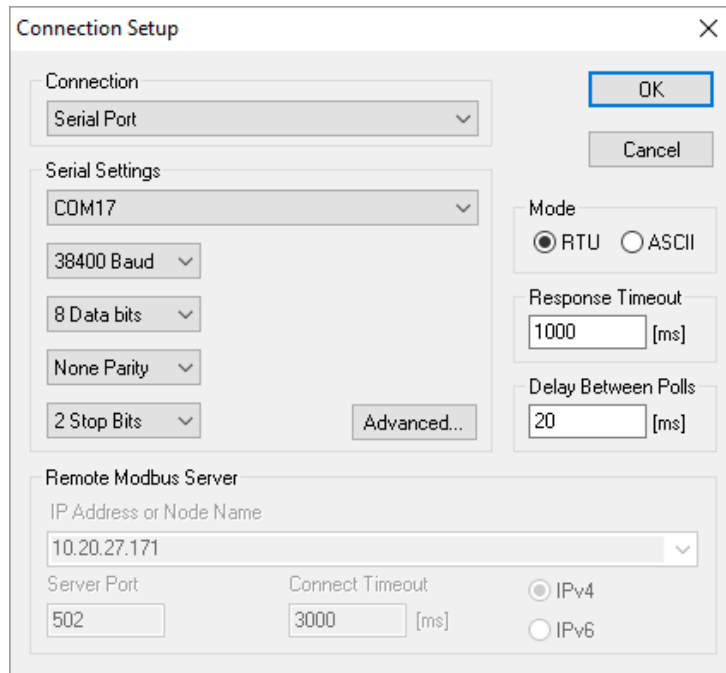
4. Start Modbus Poll. Open a *Serial Port Connection*.



Check on which COM port your RS485-USB adapter is



Pay attention to the Modbus settings on your actuator you just made



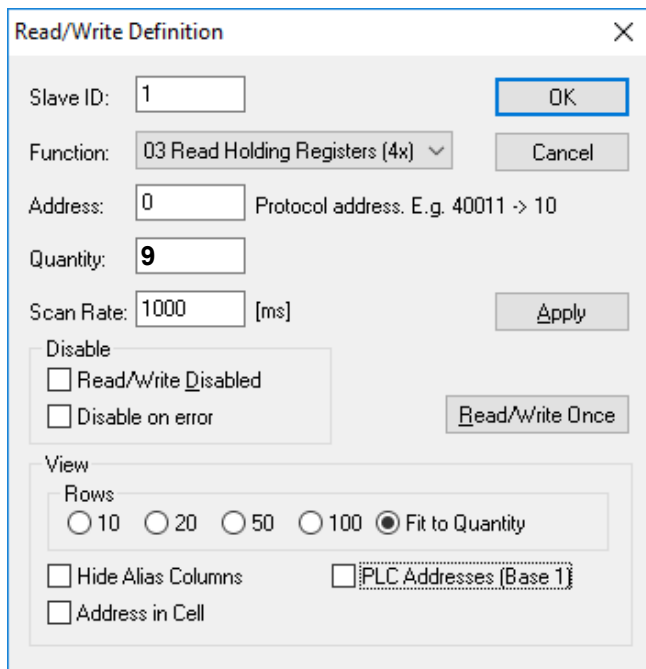
The screenshot shows the 'Connection Setup' dialog box with the following settings:

- Connection:** Serial Port
- Serial Settings:**
 - COM17
 - 38400 Baud
 - 8 Data bits
 - None Parity
 - 2 Stop Bits
- Mode:** RTU (selected), ASCII
- Response Timeout:** 1000 [ms]
- Delay Between Polls:** 20 [ms]
- Remote Modbus Server:**
 - IP Address or Node Name: 10.20.27.171
 - Server Port: 502
 - Connect Timeout: 3000 [ms]
 - IPv4 (selected), IPv6

Buttons: OK, Cancel, Advanced...

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5. Open a new window  and press the F8 key.
Make the settings according to the picture on the left.
Repeat the procedure with the settings as shown in the picture on the right.



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address: 0 Protocol address. E.g. 40011 -> 10

Quantity: 9

Scan Rate: 1000 [ms] Apply

Disable

☐ Read/Write Disabled

☐ Disable on error Read/Write Once

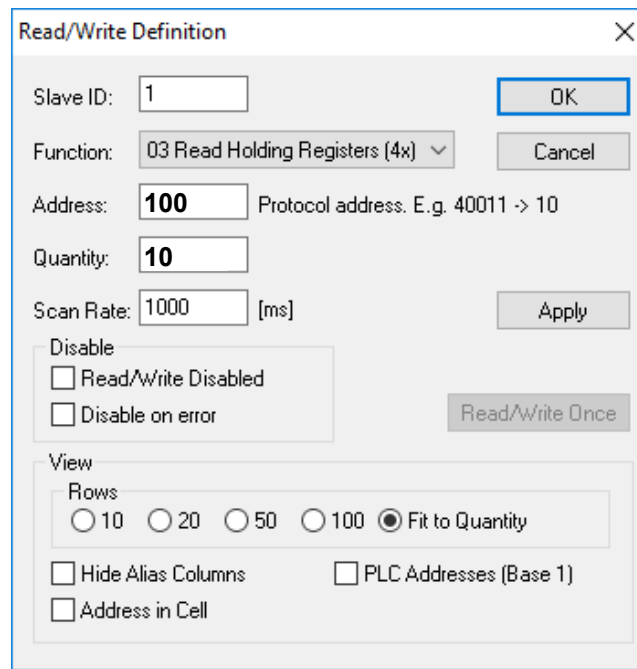
View

Rows

☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity

☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell



Read/Write Definition

Slave ID: 1 OK

Function: 03 Read Holding Registers (4x) Cancel

Address: 100 Protocol address. E.g. 40011 -> 10

Quantity: 10

Scan Rate: 1000 [ms] Apply

Disable

☐ Read/Write Disabled

☐ Disable on error Read/Write Once

View

Rows

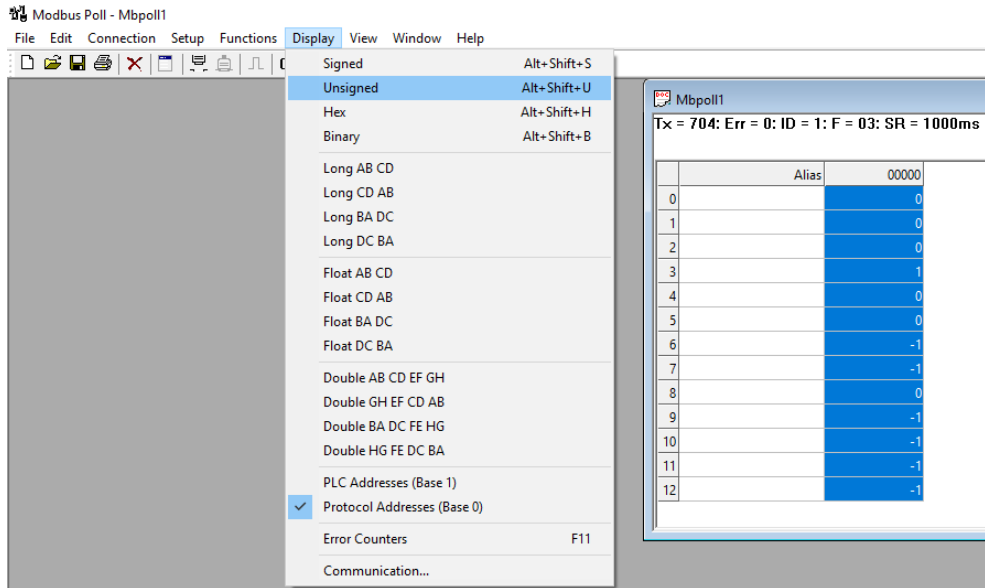
☐ 10 ☐ 20 ☐ 50 ☐ 100 ☒ Fit to Quantity

☐ Hide Alias Columns ☐ PLC Addresses (Base 1)

☐ Address in Cell

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6. Display all values as *Unsigned Integer*.



7. What is the value in register No. 4.

Value:

Interpretation:

8. Read the serial number.

Value:

Interpretation:

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9. Does the BKN has a setpoint to write?



Please see Modbus Description Register.

10. Write an override *Open*. What happens?

11. Disconnect the actuator from the BKN. What happens?

12. Read out register No. 105.

Value:

Interpretation:

13. How is register No. 105 and register No. 9 related?

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14. Connect the actuator again.

15. Try to deactivate the bus watchdog.

16. How do you reset the errors.
